

8.8 ENVIRONMENTAL ISSUES

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Pollution

With an enormous increase in human population over the last 100 years, there is a heavy demand for food, water, clothing, home, roads, transport facilities and other innumerable commodities needed by man. To meet the demand man is exerting tremendous pressure on natural resources, thereby causing **environmental pollution**.

Pollution is an undesirable change in the physical, chemical or biological characteristics of the environment due to natural causes and human activities. Any deviation from the natural composition of the environment which causes adverse effects is also described as pollution. The agents which cause pollution are called **pollutants**.

Types of pollution

More commonly pollution is classified according to the type of environment that is affected. Mainly there are three types. 1. Air Pollution 2. Water Pollution and 3. Soil Pollution.

8.8.1 Air Pollution and its control

Earth is surrounded by an envelope of air consisting of various gases which is called atmosphere. As a blanket of gases atmosphere acts as a thermal insulator and regulates the temperature on the earth by selectively absorbing U.V. rays of solar radiation. The composition of the main gases in dry air by volume is Nitrogen **78.09%**, Oxygen **20.94%**, Argon **0.93%**, Carbon dioxide **0.03%**. Life cannot exist on earth without oxygen. Air pollutants cause injury to all living organisms. They reduce growth and yield of crops. They are harmful to the respiratory system of humans and animals. Increase in the concentration of pollutants or duration of exposure increase the harmful effects on the organisms.

The major air pollutants

1. Carbon monoxide (CO)

It is produced mainly due to incomplete combustion of fossil fuels. Automobiles are a major cause of CO pollution in larger cities and towns. Automobile exhausts, fumes from factories, emissions from power plants, forest fires and even burning of fire-wood contribute to CO pollution. Haemoglobin has greater affinity for CO and so CO competitively interferes with oxygen transport. CO causes symptoms such as **headache** and **blurred vision** at lower concentrations. In higher concentrations, it leads to coma and death.

2. Carbon Dioxide (CO₂)

Carbon dioxide is the main pollutant that is leading to **global warming**. Plants utilize CO₂ for photosynthesis and all living organisms emit carbon dioxide in the process of respiration. With rapid urbanization, automobiles, aeroplanes, power plants, and other human activities that involve the burning of fossil fuels such as gasoline, carbon dioxide is turning out to be an important pollutant of concern.

3. Sulphur Dioxide (SO₂)

It is mainly produced by burning of fossil fuels. Melting of sulphur ores is another important source for SO₂ pollution. Metal smelting and other industrial processes also contribute to SO₂ pollution. Sulphur dioxide and nitrogen oxides are the major causes of **acid rains**, which cause acidification of soils, lakes and streams, and also accelerated corrosion of buildings and monuments. High concentrations of sulphur dioxide (SO₂) can result in breathing problems in asthmatic children and adults. Other effects associated with long-term exposure to sulphur dioxide, include respiratory illness, alterations in the lungs' defenses and aggravation of existing cardiovascular problems.

To control SO₂ pollution, the emissions are filtered through scrubbers*. Scrubbers are devices that are used to clean the impurities in exhaust gases .

4. Nitrogen Oxides

Nitrogen oxides are considered to be major primary pollutants. The source is mainly automobile exhaust. The air polluted by nitrogen oxides is not only harmful to humans and animals, but also dangerous for the life of plants. Nitrogen oxide pollution also results in **acid rains** and formation of **photochemical smog**. The effect of nitrogen oxides on plants include the occurrence of necrotic spots on the surface of leaves. Photosynthesis is affected in crop plants and the yield is reduced. Nitrogen oxides combine with volatile organic compounds by the action of sunlight to form secondary pollutants called **Peroxyacetyl nitrate** (PAN) which are found especially in photochemical smog*. They are powerful irritants to eyes and respiratory tract.

* Photochemical smog is a type of air pollution caused by reactions between sunlight and pollutants such as hydrocarbons and nitrogen dioxide. Photochemical smog can be extremely harmful, leading to irritations of the respiratory tract(respiratory illness) and eyes.

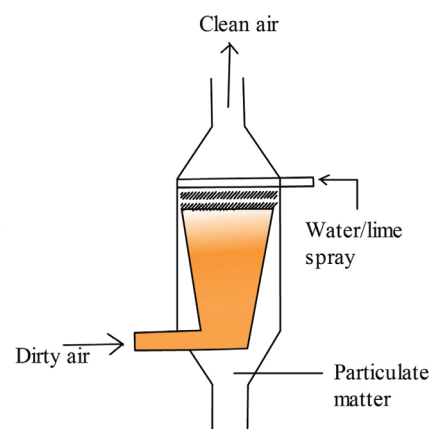


Fig. 8.22 Scrubber

*“ Scrubbers’ are air pollution control devices used to remove some particulates or gases from industrial exhausts. The term is used to describe systems that inject a dry reagent into a polluted exhaust stream to ‘wash out’/ ‘neutralise’ acid gases. Lime stone slurry is often used in scrubbers (slurry means a thin mixture of insoluble substances such as cement, dust, etc., with water)

5. Particulate matter/Aerosols

Tiny particles of solid matter suspended in a gas or liquid constitute the 'particulate matter'. 'Aerosols' refer to particles and/or liquid droplets and the gas together (**a system of colloidal particles dispersed in a gas***). Combustion of 'fossil fuels' (petrol, diesel, etc.), fly ash produced in thermal plants, forest fires, cement factories, asbestos mining and manufacturing units, spinning and ginning mills etc., are the main sources of particulate matter pollution. According to the **Central Pollution Control Board (CPCB)** particles of 2.5 micrometers or less in diameter are highly harmful to man and other air breathing organisms.

* These microscopic solids or liquid droplets that are so small and can get deep into the lungs cause decreased lung function, aggravated asthma, premature death in people with heart or lung disease, development of chronic bronchitis; irregular heartbeat, etc.

Electrostatic precipitator is a widely used 'filter' for removing particulate matter

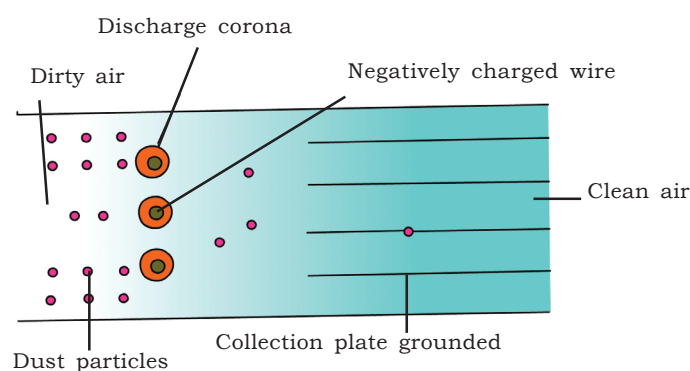


Fig 8.23 Electrostatic precipitator

from the exhaust of thermal power plants. It can remove 99% particulate matter. It has high voltage electrodes which produce a 'corona' that releases electrons. These electrons attach to dust particles making them negatively charged. These are collected by **collecting plates** which attract the charged particles. The air flowing between the plates is kept in low velocity so as to allow the dust particles to settle on plates. Thus clean air is released into the atmosphere.

Measures to control Air Pollution caused by automobiles

1. Proper maintenance of automobiles along with use of diesel or unleaded petrol.
2. Fitting '**catalytic converters**' to the automobiles having expensive metals namely platinum, palladium and rhodium as catalysts which reduce emission of poisonous gases.

Noise Pollution

In India, **Air (Prevention and control of pollution) Act** came into force in 1981. In 1987 it was amended to include noise also as an air pollutant. Undesirably high sounds constitute **noise pollution**. Sound is measured in units called decibels (dB). The human ear is sensitive to sounds ranging from **0 to 180 dB**. **0 dB** is **threshold**

limit of hearing and **120 dB** is threshold limit for sensation of pain in the ear. Any noise above **120 dB** is considered to be a noise pollution. A brief exposure to extremely high sound level, (**150 dB** or more generated by jet planes while taking-off) may damage eardrums causing permanent hearing impairment. Even long term exposure to a relatively lower level of noise of cities may also cause hearing impairment. Noise also causes auditory fatigue, anxiety, sleeplessness (**insomnia**), increased heart beat, altered breathing pattern thus causing considerable stress to humans.

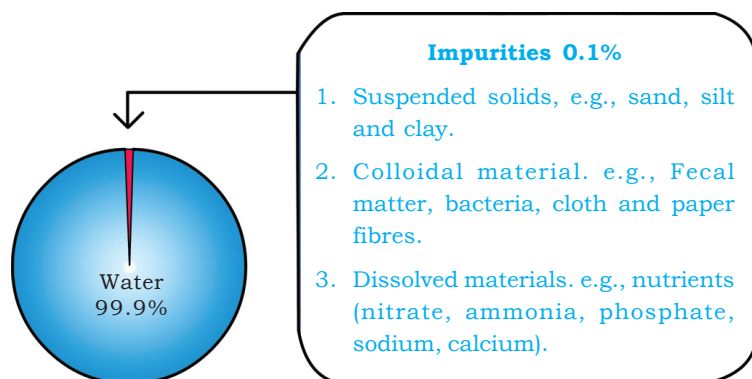
8.8.1.1 Controlling vehicular air pollution: A case study

Delhi leads the country not only as the National Capital but also in its level of air pollution. Air pollution problem in the city of Delhi reached such a serious stage that a public interest litigation (court case) was filed in the Supreme Court of India. Under the directions of the Apex Court, the Government took various measures to control air pollution. The entire fleet of public transport buses were switched over to using compressed natural gas (**CNG**) as 'fuel' instead of diesel by the end of 2002. CNG is cheaper than petrol or diesel. It is the most efficient fuel and burns almost completely or very little of it is left un-burnt in automobiles. Simultaneously parallel steps such as phasing out of old vehicles, using **unleaded** and **low sulphur petrol** and diesel, using catalytic converters in vehicles, application of stringent pollution level norms for vehicles etc., were taken by the Government. The efforts paid well and the air quality of Delhi has significantly improved.

The Government of India, through a new **Auto Fuel Policy** has laid out a roadmap to cut down vehicular pollution in Indian cities. More stringent norms for fuels are specified to steadily reduce the sulphur and the content of aromatics in petrol and diesel fuels. Euro -II norms, for example, stipulate that sulphur be controlled at **350 ppm** (parts-per-million) in diesel, and **150 ppm** in petrol. Aromatic hydrocarbons are to be contained (limited) at 42 per cent of the concerned fuel. The goal, according to the roadmap, is to reduce sulphur to **50 ppm** in petrol and diesel and bring down the level of the aromatic hydrocarbons to 35 per cent. Correspondingly/concurrently, engines of vehicles also are to be upgraded. The **Bharat Stage - III** (equivalent to **Euro-III norms**) is currently in place in many cities such as Delhi, Mumbai, Kolkata, Chennai, Bangalore, Hyderabad, Ahmedabad, Pune, Surat, Kanpur and Agra, etc. All automobiles in these major cities should have met **Euro-IV norms** by April 1, 2010.

8.8.2 Water pollution and its control

Ancient civilizations flourished along river banks. Primitive life evolved in water only. The bodies of organisms contain water to the extent of nearly 90% of their body weight. Sea water constitutes 97% of water available on the planet. Fresh water which is required for our utility constitutes only about 3%.



Inferior quality of water, caused by pollution of natural waters is a major problem world is facing today. It is posing a serious threat in developing countries and India, of course, is not an exception. Many water bodies in general and almost all the rivers in India are grossly polluted either by sewage or discharge of industrial effluents.

8.8.2.1 Ganga Action Plan

The holy river Ganga along with its 2500 kms. stretch, has been receiving for decades, millions of gallons of untreated domestic sewage from many cities and towns and also toxic industrial effluents from thousands of industries. As a result, it is becoming a **stinking dump** of various toxic substances. The water is becoming unfit for drinking and cultivation. Kanpur, Allahabad, Varanasi, Patna, and Kolkata are identified to be the main centres for causing pollution of rivers. Government of India launched a plan in 1985 with the aim of purifying the waters of river Ganga (**Ganga Action Plan**). Due to change of scope of river pollution control works, the Govt. of India renamed the programme as **National River Conservation Project / Plan**, instead of Ganga Action Plan, which is now spread in 165 towns of 17 different states.

The major water pollutants

1. Domestic Sewage

Sewage is the major source of water pollution in large cities and towns. It mainly consists of human and animal excreta and other waste materials. It is usually released into freshwater bodies or sea directly. As per the regulations the sewage has to be passed through **treatment plants** before it is released into the water courses. Only 0.1 percent of impurities from domestic sewage are making these water sources unfit for human consumption. In the treatment of sewage, solids are easy to remove.

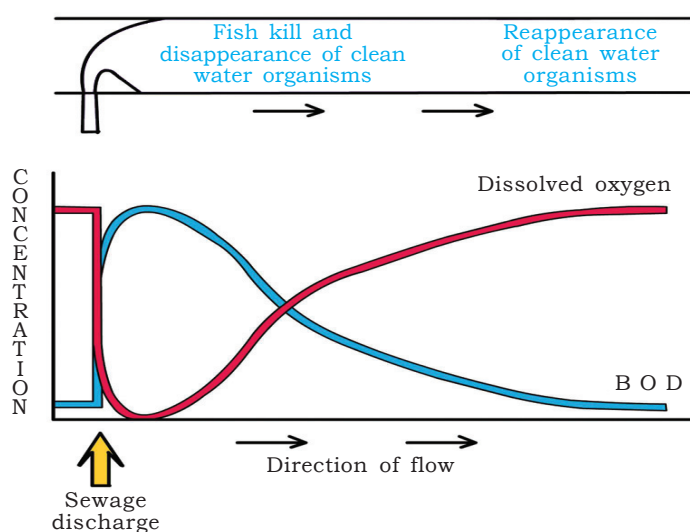


Fig. 8.23 Effect of sewage discharge on some important characteristics of a river

Removal of dissolved salts such as nitrates, phosphates and other nutrients and toxic metal ions and organic compounds is much more difficult. Domestic sewage primarily contains biodegradable organic matter, which will be readily decomposed by the action of bacteria and other microorganisms.

Biological Oxygen Demand (BOD)

BOD is a measure of the content of biologically degradable substances in sewage. The organic degradable substances are broken down by microorganisms using oxygen. The demand of oxygen is measured in terms of the oxygen consumed by microorganisms over a period of 5 days (**BOD 5**) or seven days (**BOD 7**). BOD forms an index for measuring **pollution load** in the sewage. Microorganisms involved in biodegradation of organic matter in water bodies consume a lot of oxygen, and as a result there is a sharp decline in dissolved oxygen causing death of fish and other aquatic animals.

Algal blooms

Presence of large amounts of nutrients in waters also causes excessive growth of planktonic algae and the phenomenon is commonly called 'algal blooms'. Algal blooms impart distinct colour to the water bodies and reduce the quality of water. It also causes mortality of fish. Some algae which are involved in algal blooms are toxic to human beings and animals.

Excessive growth of aquatic plants such as the *common water hyacinth* (*Eichhornia crassipes*), the world's most problematic aquatic weed which is also called '**Terror of Bengal**' causes blocks in our water ways. They grow faster than our ability to remove them. They grow abundantly in **eutrophic** water bodies (water bodies rich in nutrients) and lead to imbalance in the ecosystem dynamics of the water body.

Sewage arising from homes and hospitals may contain undesirable pathogenic microorganisms. If it is released untreated into water courses, there is a likelihood of outbreak of serious diseases, such as dysentery, typhoid, jaundice, cholera etc.,

2. Industrial Effluents

Untreated industrial effluents released into water bodies pollute most of the rivers, fresh water streams, etc., Effluents contain a wide variety of both inorganic and organic pollutants such as oils, greases, plastics, metallic wastes, suspended solids and toxins. Most of them are non-degradable. Arsenic, Cadmium, Copper, Chromium, Mercury, Zinc, and Nickel are the common heavy metals discharged from industries.

EFFECTS: Organic substances present in the water deplete the dissolved oxygen content in water by increasing the **BOD**(Biological oxygen demand) and **COD**(Chemical oxygen

demand). Most of the inorganic substances make the water unfit for drinking. Outbreaks of dysentery, typhoid, jaundice, cholera etc., are caused by sewage pollution.

Biomagnification

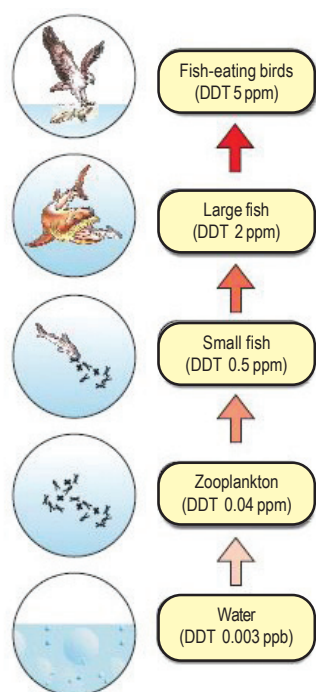


Fig. 8.24 Biomagnification of DDT in an aquatic food chain

Increase in the concentration of the pollutant or toxicant at successive trophic levels in an aquatic food chain is called **Biological Magnification** or **Bio-magnification**. This happens in the instances where a toxic substance accumulated by an organism is not metabolized or excreted and thus passes on to the next higher trophic level. This phenomenon is well known regarding DDT and mercury pollution.

As shown in the Fig. 8.24, the concentration of **DDT** is increased at successive trophic levels starting at a very low concentration of **0.003 ppb** (ppb = parts per billion) in water, which increased to a high concentration of **5 ppm** (ppm = parts per million) in fish-eating birds, through bio-magnification (It can reach maximum upto 25 ppm). High concentrations of DDT disturb **calcium metabolism in birds**, which causes thinning of egg shell and their premature breaking, eventually causing decline in bird populations.

Eutrophication

Natural **ageing** of a lake by nutrient enrichment of its water is known as **eutrophication**. In a young lake, the water is cold and clear, supporting little life. Gradually nutrients such as nitrates and phosphates are carried into the lake via streams, in course of time. This encourages the growth of aquatic algae and other plants. Consequently the animal life proliferates, and organic matter gets deposited on the bottom of the lake. Over centuries, as silt and organic debris piles up, the lake grows shallower and warmer. As a result, the aquatic organisms thriving in the cold environment are gradually replaced by warm-water organisms. Marsh plants appear by taking root in the shallow regions of the lake. Eventually, the lake gives way to large masses of floating plants (bog) and finally converted into land.

A **bog** is a wet, spongy, acidic ground composed of decayed vegetable matter.

Depending upon the climatic conditions, size of the lake and other factors, the natural ageing of a lake may span thousands of years. However, pollutants from human activity (**anthropogenic**) radically accelerate the aging process. This phenomenon is called '**Cultural or Accelerated eutrophication**'.

During the past century, lakes in many parts of the earth have been severely eutrophied by sewage, agricultural and industrial wastes. The prime contaminants

are nitrates and phosphates, which are the '**chief plant nutrients**'. The dissolved oxygen which is vital to other aquatic life is depleted. At the same time, other pollutants flowing into the lake may poison the whole population of fish, whose decomposing remains further deplete the dissolved oxygen content in the water.

Thermal pollution

Water is used as a coolant in Thermal power plants and other industries. Hot water flowing out of industries also constitute an important category of pollutants. Thermal waste water eliminates sensitive organisms (**stenothermal** organisms such as fish- especially the juveniles) downstream and may enhance the growth of plants and fish in extremely cold areas but, only after causing damage to the indigenous flora and fauna.

8.8.2.2 A Case Study Of Integrated Waste Water Treatment

Wastewater including sewage can be treated in an integrated manner, by combining both artificial and natural processes. A good example for such an initiative comes from the town of **Arcata**, situated along the northern coast of **California**. In collaboration with biologists from the Humboldt State University, the people of the town created an integrated waste water treatment process within a natural system. The cleaning of waste water is done in two stages.

1. In the **first stage**, the conventional sedimentation, filtering and chlorination treatments are given. Even after this treatment, the waste water contains lots of harmful pollutants such as dissolved heavy metals.
2. In the **second stage**, the water is discharged into a series of six connected marshes over 60 hectares of marshland seeded with appropriate plants, algae, fungi and bacteria as suggested by biologists. These flora of the marshland **neutralise, absorb and assimilate the pollutants**. As the water flows through the marshes, it is purified in a natural way and can be used for cultivation or other purposes.

The marshes also constitute a **sanctuary**, with a high level of **biodiversity** in the form of fishes, animals and birds that inhabit the area. A citizens' group called **Friends of the Arcata Marsh** (FOAM) are responsible for the upkeep and safeguarding of the wonderful project.

Ecological Sanitation - 'Ecosan Toilets'

Generally it is assumed that removal of wastes requires water, which means **creation of sewage**. If water is not used to dispose off human waste like excreta, and if one didn't have to flush the toilet after its use, a large amount of water can be saved. This is already a reality. Ecological sanitation is a sustainable system for handling human excreta, using '**Dry composting toilets**'. This is a practical, hygienic, efficient and cost-effective solution to human waste disposal. The key point to note here is that,

with this composting method, human excreta can be recycled into a resource (as natural fertiliser), which reduces the need for chemical fertilizers. 'EcoSan' toilets are in use in many parts of Kerala and Sri Lanka.

8.8.3 Soil Pollution (Land Pollution)

Soil or land pollution is mainly due to solid wastes and agro-chemicals such as fertilizers and pesticides.

Solid Wastes

Any thing (substance/material/articles/goods) that is thrown out as waste in solid form is referred to as **solid waste**. Municipal solid wastes are wastes from homes, offices, institutions, shops, hotels, restaurants etc., in towns and cities.

The municipal solid wastes generally consist of paper, food wastes, plastics, glass, metals, rubber, leather, textile, etc., The wastes are **burnt to reduce the volume** of the wastes. But generally wastes are not completely burnt and left as open dumps which often serve as the breeding grounds for rats and flies. As the substitute for open-burning dumps, **sanitary landfills** are adopted. In a sanitary landfill, wastes are dumped in a depression or trench after compaction, and covered with dirt everyday. There is a danger of seepage of chemicals and pollutants from these landfills, which may contaminate the underground water resources.

The best solution is to develop awareness in the society on these environmental issues. All wastes that we generate can be categorized into three types (a) **bio-degradable**, (b) **recyclable** and (c) **non-biodegradable**. It is important that all garbage generated should be sorted out category wise. The reusable or recyclable material has to be separated out and utilised. (Rag-pickers in the streets are doing a great job of separation of materials for recycling.) The biodegradable materials can be put into deep pits in the ground and be left for natural breakdown. The remaining non-biodegradable waste left over is to be disposed off properly.

The prime goal should be to reduce our garbage generation. But we are increasing the use of non-biodegradable products. We are packaging products of our daily use such as milk and water also in polythene bags. In cities and towns, many purchased things are packed in **polystyrene** and **plastic** packets. Thus we are contributing heavily to environmental pollution. State Governments across the country are trying to educate people on the reduction in use of plastics and use of **eco-friendly packaging**. We can do our bit by using carry-bags made of cloth or other natural fibres when we go for shopping and by refusing polythene bags.

i) Hospital Wastes

Hospitals generate hazardous wastes that contain disinfectants, harmful chemicals and also pathogenic micro-organisms. Such wastes also require careful treatment and

disposal. The use of **incinerators** (to burn wastes) is essential for disposal of hospital waste.

ii) Electronic wastes (e-wastes):

Irreparable computers and other electronic goods constitute the modern day pollutants called electronic wastes (e-wastes). E-wastes are buried in landfills or incinerated. Over half of the e-wastes generated in the developed world are exported to developing countries, mainly to China, India and Pakistan, where metals like copper, iron, silicon, nickel and gold are recovered during recycling process. Unlike developed countries, which have specifically built facilities for recycling of e-wastes, recycling in developing countries often involves manual participation thus exposing workers to toxic substances present in e-wastes. Eventually recycling is the only solution for the treatment of e-wastes provided it is carried out in an **environmental friendly** manner.

8.8.3.1 A Case Study Of Remedy For Plastic Waste

A Bangalore based plastic sack manufacturer **Ahmed Khan**, found an ideal solution to the ever-increasing problem of accumulating plastic waste. He is in the field of producing plastic sacks for more than 20 years. A few years ago, he realised the threat posed by plastic waste. As a remedy, his company developed a fine powder of recycled plastic mixture named **polyblend**. Polyblend is mixed with the **bitumen** that is used to lay roads. In collaboration with R.V.College of Engineering and the Bangalore City Corporation, Ahmed Khan proved that the life of roads laid with **polyblend** and **bitumen** mixture is tripled. It is observed that polyblend enhanced the bitumen's water repellant properties, and helped increase its road life. The raw material for creating polyblend is any plastic film waste. The cost of plastic waste given to rag pickers rose from Rs. 0.40 per kg to Rs.6 as was offered by Khan. Using Khan's technique, several roads have already been laid in Bangalore. Khan's company will soon be running short of plastic waste in Bangalore, to produce Polyblend. Thanks to innovations like Polyblend, we could find at least one way to overcome the problem of plastic wastes.

iii) Agro-chemicals and Their Effects

In the wake of the **Green Revolution**, use of inorganic fertilisers and pesticides has increased many times, for enhancing crop production. Pesticides, herbicides, fungicides, etc., are being increasingly used. They are also toxic to non-target organisms such as earthworms, nitrogen fixing bacteria, etc., that are important components of soil eco-system. Moreover due to bio-magnification, the harmful chemicals pose a great threat to human health. Indiscriminate use of fertilizers will lead to increased drain of nutrients into the nearby aquatic ecosystems causing eutrophication and the consequent effects.

8.8.3.2 Case Study of organic farming

Integrated organic farming is a **zero-waste** procedure, where recycling of waste products is efficiently carried out. Wastes originated from one process are used as nutrients for other processes. This allows the maximum utilisation of resource and increases the efficiency of production. A method practiced by **Ramesh Chandra Dagar**, a farmer in Sonipat, Haryana, is a very good example for this. He integrated bee-keeping, dairy management, water harvesting, composting and agriculture in a chain of processes. All these processes support one another and allow an extremely economical and sustainable venture. Crop waste and cattle excreta (dung) are used to create **compost**, which can be used as a natural fertilizer. Natural **bio-gas** generated in the process can be used for meeting the energy needs of the farm. Enthusiastic about spreading information and helping in the practice of integrated organic farming, Dagar has created the **Haryana Kisan Welfare Club**.

iv) Radioactive wastes

Initially, nuclear energy was considered a non-polluting way for generating electricity. Later on, it was realised that the use of nuclear energy has two very serious inherent problems. The first is **accidental leakages**, as occurred in the **Three Mile Island (USA)** and **Chernobyl (Russia)** and the second is the **safe disposal** of radioactive wastes.

Radiation, that is released from nuclear waste is extremely dangerous to biological organisms, because it induces mutations. Exposure to high doses of nuclear radiation can lead to cancers (e.g. leukemia)., Therefore, nuclear waste is an extremely **potent pollutant** and has to be dealt with utmost caution. Storage of nuclear wastes should be done in suitably shielded containers and buried deep in the soil or oceans (about 500 meters). Even when done so, geological upheavals can bring them up, some day and cause radiation.

8.8.4 Green house effect and global warming

The term '**Greenhouse effect**' has been derived from a phenomenon that occurs in a 'greenhouse'. Greenhouse is a small glass house and is used for growing plants especially during winter. In a greenhouse the glass panel allows the passage of light into it, but does not allow heat to escape (as it is reflected back). Therefore, the greenhouse warms up, very much like inside a car that has been parked in the sun for a few hours.

The greenhouse effect is a naturally occurring phenomenon that is responsible for heating of the Earth's surface and atmosphere. It would be surprising to know that without greenhouse effect the average temperature of the Earth's surface would have been a chilly -18°C rather than the present average of 15°C .

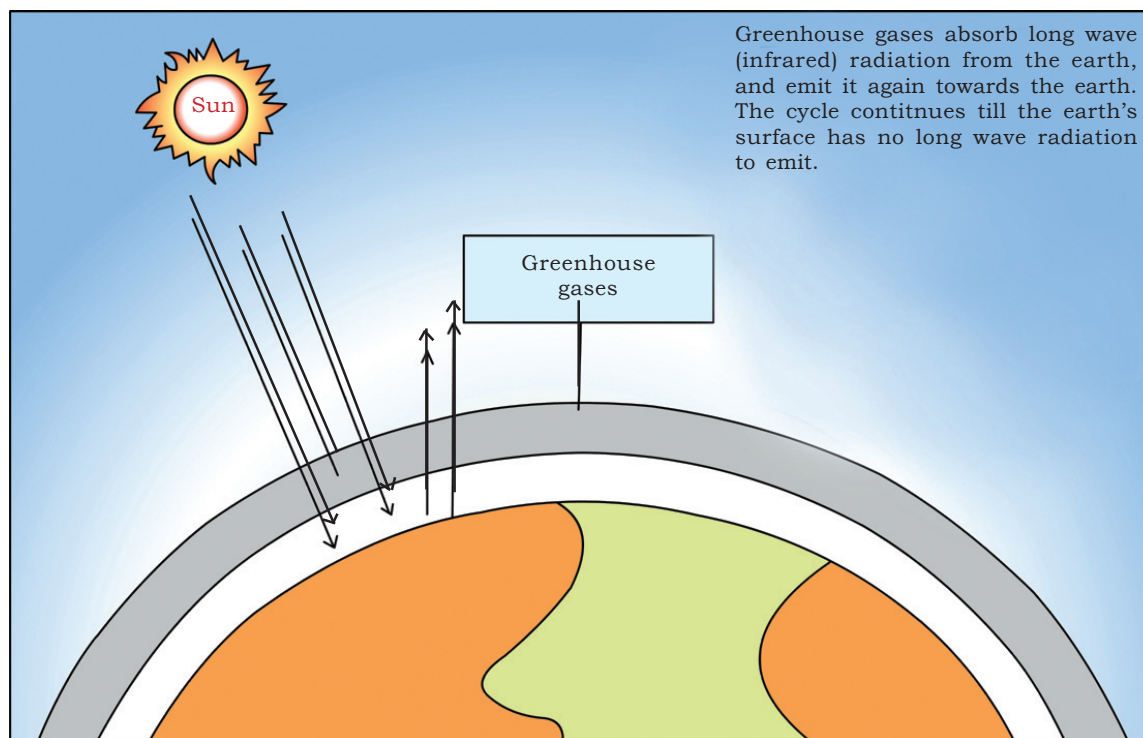


Fig. 8.25 Sunlight energy at the outermost atmosphere

When sunlight reaches the outermost layer of the atmosphere, clouds and gases reflect about one-fourth of the incoming solar radiation, and absorb some of it. Almost half of the incoming solar radiation falls on the Earth's surface and heats it up, while a small proportion is reflected back. The Earth's surface reflects heat in the form of infrared radiation but part of this does not escape into space as atmospheric gases (e.g., carbon dioxide, methane, etc.) absorb a major fraction of it. The molecules of these gases radiate heat energy, and a major part of which again comes back to the Earth's surface, thus **heating it up once again**. The above-mentioned gases – **carbon dioxide** and **methane** are commonly known as **greenhouse gases**, because they are responsible for the **greenhouse effect**.

Increase in the level of greenhouse gases has led to considerable heating of the Earth leading to **global warming**. During the past century, the temperature of the Earth has increased by 0.6 °C, most of it during the last three decades. Scientists believe that this rise in temperature is leading to severe changes in the environment. Global warming is causing climatic changes (e.g. such as **El Nino effect***) and is also responsible for the melting of polar ice caps and other snow caps of mountains such as the Himalayas. Over many years, this will result in a rise in sea levels, all over the world, that can submerge many coastal areas in the world. The total spectrum of

* El nino is unusual weather phenomena caused due to rise in temperature of the surface waters of the Pacific ocean in different parts of the world. Its effects are experienced all over the world and more so in the South American countries

changes that global warming can bring about is a subject that is still under active research.

Global warming – Control measures

1. The measures include cutting down use of fossil fuels.
2. Improving efficiency of energy usage.
3. Planting of trees, and avoiding deforestation.
4. Slowing down the growth of human population.

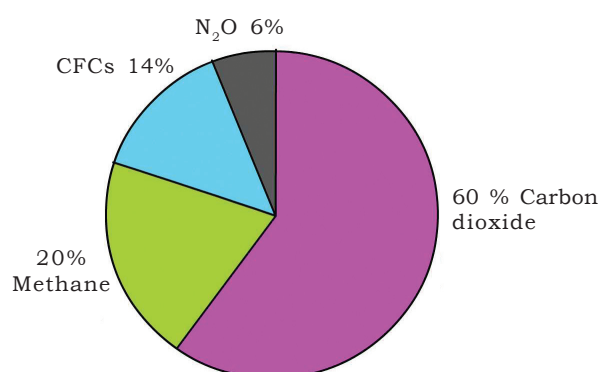


Fig 8.26 Relative contribution of various green house gases to total global warming

Kyoto Protocol

International initiatives are also being taken to reduce the emission of greenhouse gases into the atmosphere. To save earth from the dangers of 'global warming', an international conference was held at Kyoto, Japan (1997). As of September 2011, 191 countries have signed and ratified the protocol. As per this resolution it is decided to achieve the goal of reducing **green house emissions** into the atmosphere by 2008-12 to the levels of 1990.

8.8.5 Ozone depletion in the

stratosphere

Ozone, formed in the lower atmosphere (troposphere) is known as '**bad ozone**' which harms plants and animals. Ozone found in the upper part of the atmosphere called the stratosphere is known as '**good ozone**', and it acts as an efficient **shield** by absorbing ultraviolet radiation from the sun. UV rays are highly injurious to living organisms. DNA and proteins of living organisms preferentially absorb UV radiation, and its high energy breaks the chemical bonds within these molecules. The thickness/ columnar density of the ozone in a **column of air** is measured in terms of **Dobson units** (DU). Ozone gas is continuously formed by the action of UV rays on molecular oxygen and is unstable. It is degraded into molecular oxygen in the stratosphere. There will be generally a balance between production and degradation of ozone in the stratosphere. Of late, the balance has been disrupted due to enhancement of ozone degradation by chlorofluorocarbons (**CFCs**). CFCs are widely used as refrigerants and coolants. CFCs discharged into the lower strata of the atmosphere, move upward and reach the stratosphere. In the stratosphere, Cl atoms are released from CFC molecules by the action of UV rays. These Cl atoms act as catalysts and degrade ozone molecules and release **molecular oxygen**. As the Cl atoms are not consumed in this reaction, they continue to degrade ozone molecules. Hence, whatever CFCs are added to the stratosphere, they have permanent and continuing effects on the levels of ozone. The

depletion of ozone is particularly marked over the **Antarctic region**. This has resulted in the formation of a large area of thinned ozone layer, commonly called as the '**ozone hole**' (Figure).

UV radiation with wavelengths shorter than that of UV-B, are almost completely absorbed by Earth's atmosphere, provided that the ozone layer is intact. But **UV-B damages DNA and may induce mutations**. It causes ageing of skin, damage to skin cells resulting in **skin cancers**. In human eye, cornea absorbs **UV-B** radiation, and a high dose of **UV-B** causes inflammation of **cornea**, called **snow-blindness** formation of cataract, etc., Such exposure may permanently damage the cornea.

Montreal Protocol

Understanding the deleterious effects of depletion of ozone, an international treaty, known as the **Montreal Protocol**, was signed at Montreal (Canada) in **1987** (became **effective in 1989**) to control the emission of pollutants/ substances responsible for ozone depletion. Subsequently many more efforts have been made and protocols have laid down definite roadmaps, separately for developed and developing countries, for reducing the emission of CFCs and other ozone depleting chemicals.

8.8.6 Degradation by improper resource utilisation and maintenance

The degradation of natural resources can occur, not just by the action of pollutants but also by improper resource utilisation practices.

- i) **Soil erosion and desertification**: The development of the fertile **top-soil** takes centuries. But, it can be removed very easily due to human activities like over-cultivation, unrestricted grazing, deforestation and poor irrigation practices, resulting in the formation of arid patches of land. When large barren patches extend and meet over time, a desert is created. Internationally, it has been recognised that **desertification** is a major problem nowadays, particularly due to increased urbanisation.
- ii) **Water-logging and soil salinity**: Irrigation without proper drainage of water leads to water-logging in the soil. Besides affecting the crops, water-logging draws salt to the surface of the soil (**salinisation of the top soil**). The salt then is deposited as a thin crust on the land surface or starts collecting at the roots of the plants.

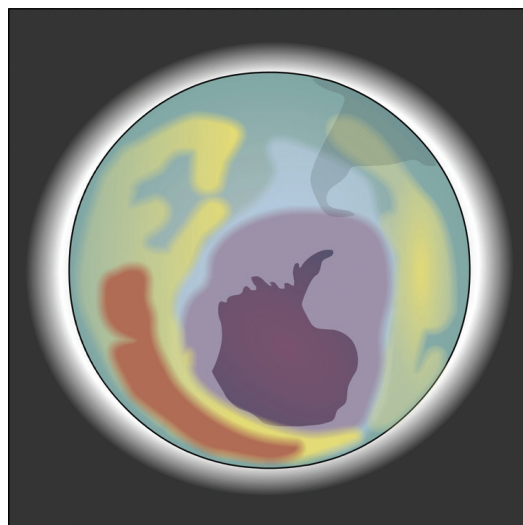


Fig. 8.27 Ozone hole is the area above Antarctica, shown in purple colour, where the ozone layer is the thinnest. Ozone thickness is given in dobson units (see carefully the scale shown in colour violet or red). The ozone hole over Antarctica develops each year between late August and early October. *Curtesy: NASA*

This increased salt content is inimical (unfavourable) to the growth of crops and is extremely damaging to agriculture. Water-logging and soil salinity are some of the problems that have come in the wake of the Green Revolution.

iii) Deforestation: Deforestation is the conversion of forested areas to non-forested ones. According to an estimate, almost forty per cent forests have been lost in the tropics, compared to only one per cent in the temperate regions. At the beginning of the twentieth century, forests covered about **30** per cent of the land of India. By the end of the century, it shrunk to **19.4** per cent, whereas the **National Forest Policy (1988)** of India has recommended **33** percent forest cover for the plains and 67 per cent for the hills as ideal or desirable.

A number of human activities contribute to deforestation. One of the major reasons is the conversion of forest to agricultural land so as to feed the growing human population. Trees are axed for timber, firewood, cattle ranching and for several other purposes. **Slash** (cut) and **burn agriculture**, commonly called as **Jhum cultivation** in the north-eastern states of India and **podu** in other parts, has also contributed to deforestation. In slash and burn agriculture, the farmers cut down the trees of the forest and burn the plant remains. The ash serves as a fertiliser and the land is then used for farming or grazing cattle. After cultivation, the area is abandoned for several years to allow its recovery. The farmers then move on to other areas and repeat this process.

Consequences of deforestation

- 1) Enhanced carbon dioxide concentration in the atmosphere because trees that could hold a lot of carbon in their biomass (use CO_2 for photosynthesis) are lost with deforestation.
- 2) Loss of biodiversity due to habitat destruction.
- 3) Disturbed hydrological cycle.
- 4) Soil erosion.
- 5) Desertification in extreme cases.

Reforestation

Reforestation is the process of restoring a forest that once existed but was destroyed at some point of time in the past. Reforestation may occur naturally in a deforested area. However, we can speed it up by planting trees(saplings) with due consideration to biodiversity that existed earlier in that area.

8.8.6.1 Case Study of People's Participation in the Conservation of Forests

Amrita Devi Bishnoi wildlife protection Award

People's participation in 'protecting forests' has a long history in India. In 1731, the king of Jodhpur in Rajasthan asked one of his ministers to arrange wood for constructing a new palace. The minister and workers went to a forest near a village, inhabited by Bishnois, to cut down trees. The **Bishnoi community** is known for its peaceful co-existence with nature. The effort to cut down trees by the king was thwarted by the Bishnois. A Bishnoi woman **Amrita Devi** showed exemplary courage by hugging a tree and daring king's men to cut her first before cutting the tree. The tree mattered much more to her than her own life. Sadly, the king's men did not heed to her pleas, and cut down the tree along with Amrita Devi. Her three daughters and hundreds of other Bishnois followed her, and thus lost their lives in their effort to save trees. Nowhere in history do we find a commitment of this magnitude when human beings sacrificed their lives for the cause of the environment. The Government of India has recently instituted the "**Amrita Devi Bishnoi Wildlife Protection Award**" for individuals or communities from rural areas that have shown extraordinary courage and dedication in protecting wildlife.

The Chipko Movement of Garhwal Himalayas

In 1974, local women showed enormous bravery in protecting trees from the axe of contractors by hugging the trees. People all over the world have acclaimed the **Chipko movement**. Realising the significance of participation by local communities the Government of India in 1980s has introduced the concept of **Joint Forest Management (JFM)** so as to work closely with the local communities for protecting and managing forests. In return for their services to the forest, the communities get the benefit of various forest products (e.g., fruits, gum, rubber, medicine, etc.).

Indian Government passed Environment (Protection) Act in 1986; Water (Prevention and Control of Pollution) Act in 1974.

GLOSSARY

Autecology: ecology of individual species

Basking: Exposing the body to sun light, to gain temperature.

Benthos: It refers to all the attached, creeping or burrowing organisms that inhabit the bottom of rivers, lakes and sea.

Biomass: The total mass of living material within a specified area at a given time.

Blubber: It is a specialized subcutaneous layer of fat found only in marine mammals. It is almost continuous across the body of marine mammals but absent on appendages.

Brackish water: An intermediate zone between freshwater and marine water.

Camouflage Concealing coloration (e.g., melanism) and morphology (e.g. stick insects) as defence against predation

Chemoautotrophs: These comprise bacteria that obtain energy from the oxidation of simple inorganic compounds and can use the energy released to assimilate CO₂ and transfer the energy into organic compounds. E.g. *Thiobacillus* species.

Climate: The climate of an area can be described by its mean values of temperature, rainfall, wind speed.

Community: The total living biotic component of an ecosystem, including plants, animals and microbes.

Competitive exclusion: It is often known as Gause principle.

Cyclomorphosis: Cyclic change in phenotype, such as seasonal changes in morphology, particularly conspicuous among cladoceran crustacean and rotifers.

Detritus: Non living organic matter. Usually refers to particulate matter to that of plant rather than animal origin. E.g. leaf litter.

Diapause: It is a condition of arrested growth or reproductive development common in many organisms, particularly insects, during unfavourable conditions.

Dimictic Lake: The lake that undergoes two periods of complete vertical mixing, usually in the spring and the fall. During the summer dimictic lakes are thermally stratified.

Edaphic factors: The physical, chemical and biotic characteristics of the soil that influence plant growth and distribution.

Estuary: It is a place where river joins the sea. The water in an estuary is subjected to seasonal variations in salinity. The water is called brackish water. The animals living there are euryhaline.

Gemmules: These are the internal buds that appear in the asexual reproduction of sponges. Gemmules are made up of amoebocytes and covered by a layer of spicules, and can survive in unfavourable conditions.

Leaching: Removal of soluble components by flowing water from soil.

Mycorrhizae: The symbiotic association of mycelium of a fungus with the roots of a seed plant.

Osmotrophic nutrition: Intake of pre-digested food material through the body surface.

Pedonic forms: The organisms which depend on substratum (or support) in an aquatic ecosystem.

Periphyton: The communities of tiny organisms like protozoa, insect larvae, snails that live on the surfaces of aquatic plants.

Savannas: Grassland region with scattered trees in subtropical and tropical regions.

Standing crop: The mass of vegetation in a given area at one particular time. Although most of often applied to plant material, the term includes animal biomass.

Denitrificaion: Denitrification is a microbially facilitated process of nitrate reduction that may ultimately produce molecular nitrogen (N_2) through a series of intermediate gaseous nitrogen oxide products.

Decomposer: Decomposers (or saprotrophs) are organisms that break down dead or decaying organisms, and in doing so carry out the natural process of decomposition.

Sedimentary: Formed by the accumulation and consolidation of mineral and organic fragments that have been deposited by water, ice, or wind.

Asymptote: A line which approaches nearer to some curve than assignable distance, but, though infinitely extended, would never meet it.

Chemo-autotroph: An organism (typically a bacterium or a protozoan) that obtains energy through chemical process, which is by the oxidation of electron donating molecules from the environment, rather than by photosynthesis.

Mortality: Death rate or the number of individuals that died in a population in a unit time.

Natality: Birth rate or the number of individuals produced in a population in unit time.

Acid rains: Acid rain is a rain or any other form of precipitation that is unusually acidic.

Algal bloom: An algal bloom is a rapid increase or accumulation in the population of algae (typically microscopic) in an aquatic system.

Biodegradable: Capable of being broken down especially into harmless products by the action of living things (as microorganisms).

Biological Oxygen Demand (BOD): The amount of dissolved oxygen needed by aerobic biological organisms in a body of water to break down organic material present in a given water sample at certain temperature over a specific time period.

Chemical Oxygen Demand (COD): A test procedure based on the chemical decomposition of organic and inorganic contaminants dissolved or suspended in water.

Chloro fluoro carbons (CFC): Any of various halocarbon compounds consisting of carbon, hydrogen, chlorine, and fluorine, once used widely as aerosol propellants and refrigerants. Chlorofluorocarbons are believed to cause depletion of the atmospheric ozone layer.

Deforestation: The removal of a forest or stand of trees where the land is thereafter converted to a non-forest use.

Eutrophication: A process whereby water bodies, such as lakes, estuaries, or slow-moving streams receive excess nutrients that stimulate excessive plant growth (algae, periphyton attached algae, and nuisance plants and weeds).

Fungicides: Substance or preparation, as a spray or dust, used for destroying fungi.

Herbicides: Chemicals used to kill unwanted weeds.

Incinerator: a furnace or apparatus for burning trash, garbage, etc., to ashes.

Land fills: Landfill is a carefully engineered depression in the ground (or built on top of the ground, resembling a football stadium) into which wastes are dumped.

Pesticides: Chemicals used to kill pests, especially insects.

Photochemical smog: A type of air pollution produced when sunlight acts upon motor vehicle exhaust gases to form harmful substances such as ozone (O_3), aldehydes and peroxyacetylnitrate (PAN).

Polyblend: Physical mixture of two or more polymers. Such blends usually yield products with favorable properties of both components.

Scrubber: Scrubbers are commonly used to eliminate potentially harmful dust and pollutants from exhausts. In scrubbers, a liquid, in general water added with active chemicals is sprayed in to the air flow. Aerosol and gaseous pollutants in the air stream are removed by either absorption or chemical reactions with the water solution.

Sewage: Domestic waste water containing various solid and liquid waste materials including human excreta.

Soil erosion: The washing away of soil by the flow of water or wind

Thermal Pollution: Water Pollution caused by hot water coming out from industries, thermal power plants etc., which is harmful to aquatic organisms.

Ultraviolet-B (UV-B): One of the three types of invisible light rays (together with ultraviolet-A and ultraviolet-C) given off by the sun. Although ultraviolet-C is the most dangerous type of ultraviolet light in terms of its potential to harm life on earth, it cannot penetrate earth's protective ozone layer. Therefore, it poses no threat to human, animal or plant life on earth as long as ozone layer is intact. Ultraviolet-A and ultraviolet-B, on the other hand, do penetrate the ozone layer in attenuated form and reach the surface of the planet. UV-A rays cause cells to age and can cause some damage to cells' DNA. They are linked to long-term skin damage such as wrinkles, but are also thought to play a role in some skin cancers. UVB rays can cause direct damage to the DNA, and are the main rays that cause sunburns. They are also believed to cause most skin cancers.

VERY SHORT ANSWER TYPE QUESTIONS

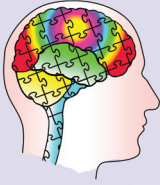
1. Define the term ecology and its branches.
2. Define a community?
3. What is an ecosystem?
4. What is a biome? Name any two biomes you studied.
5. Explain the difference between the 'niche' of an organism and its 'habitat'.
6. How does your body solve the problem of altitude sickness, when you ascend tall mountains?
7. What is the effect of light on body pigmentation?
8. Distinguish the terms phototaxis and photokinesis.
9. What is photoperiodism?
10. Mention the advantages of some UV rays to us.
11. What is cyclomorphosis? Explain its importance in *Daphnia*.
12. What are 'regulators'?
13. What are 'conformers'?
14. Define commensalism. Give one example.
15. Define mutualism. Give one example.
16. What is camouflage? Give its significance.
17. What is Gause's principle? When does it applicable?
18. Distinguish between lotic and lentic habitats.
19. Distinguish between phytoplankton and zooplankton.
20. Distinguish between neuston and nekton.
21. Explain the process of 'leaching'?
22. What is PAR?
23. What is the percentage of PAR, in the incident solar radiation.
24. Explain the terms GPP, NPP.
25. Distinguish between upright and inverted ecological pyramids.
26. Distinguish between primary and secondary productivity.
27. Which air pollutants are chiefly responsible for acid rains?
28. What is BOD?
29. What is biological magnification?
30. Why are incinerators used in hospitals?

SHORT ANSWER TYPE QUESTIONS

1. What is summer stratification? Explain.
2. What is the significance of stratification in lakes?
3. Explain van't Hoff rule.
4. Unlike mammals the reptiles cannot tolerate environmental fluctuations in temperature. How do they adapt to survive in desert conditions?
5. How do terrestrial animals protect themselves from the danger of dehydration of bodies?
6. How do marine animals adapt to hypertonic seawater?
7. Distinguish between euryhaline and stenohaline animals.
8. An orchid plant is growing on the branch of mango tree. How do you describe this interaction between the orchid and mango tree?
9. Predation has a significant role in maintaining of species diversity-discuss.
10. Discuss competitive release.
11. Explain brood parasitism with a suitable example.
12. How do predators act as biological control?
13. Explain the different types of aquatic ecosystems.
14. Explain the terms saprotroph, detritivores, and mineralizers.
15. Define decomposition and describe the process and products of decomposition.
16. What is primary productivity? Give a brief description of the factors that affect primary productivity.
17. Define ecological pyramids and describe with examples, pyramids of numbers and biomass.
18. What are the deleterious effects of depletion of ozone in the stratosphere?
19. Describe 'Green House Effect'?
20. Write critical notes on the following:
 - a) Eutrophication
 - b) Biological magnification
 - c) Ground water depletion and ways for its replenishment

**LONG ANSWER TYPE
QUESTIONS**

1. Write an essay on temperature as an ecological factor.
2. Write an essay on water as an ecological factor.
3. Describe lake as an ecosystem giving examples for the various zones and the biotic components in it.
4. Describe different types of food chains that exist in an ecosystem.
5. Give an account of flow of energy in an ecosystem.
6. List out the major air pollutants and describe their effects on human beings ?
7. What are the causes of water pollution and suggest measures for control of water pollution ?



For ignited minds

- NOT FOR EVALUATION

1. All types of UV rays are not useless to man. How do you defend the statement?
2. Elephant is a large sized animal and an herbivore. How do you justify it?
3. Lakes in temperate countries are much better places for aquatic life than the tropical lakes. Do you agree with above statement? Explain.
4. Even though different kinds of plants live in a lake, there is a sort of 'war' between some species, which is not visible to the naked eye. Justify the comment giving an example.
5. With reference to representation of biomasses in food pyramids, what happens when algae 'bloom' and after a month or two the algae start dying, as they have completed their life cycles.
6. "All photosynthetic organisms are autotrophs, but all autotrophs are not photosynthetic." How do you substantiate this statement.
7. Based on the laws of thermodynamics, guess if bioluminescence influences the 'energy homeostasis' in the body. (Clue: bioluminescence is a catabolic process, in a way.)
8. If the body mass of a tertiary carnivore is 10 KJ, and if the photosynthetic organisms which provide food for it utilize 20% of their produce in their catabolism, what will be the GPP taking into consideration the 10% law.
9. Why do marine birds excrete salts through nasal passages?
10. Can a fungus become a part of the predator chain? If so, can you give an example?

References

- ❖ Biological sciences –D.J Taylor ,N.P.O Green and G.W Stout.
- ❖ Integrated principles of zoology-Hickman, Roberts and Larson
- ❖ Environmental science towards a sustainable future-Richard t Wright, Dorothy F.Boorse
- ❖ Environmental science-Y. Anjaneyulu
- ❖ Biodiversity principles and conservation-Kumar, Asija.
- ❖ Zoology Phyla series - R .L. Kotpal, Rastogi Publications, Shivaji Road, Meerut.
- ❖ Invertebrate Zoology – E.L. Jordan and P.S. Verma, S. Chand & company Ltd. Ram Nagar, New Delhi.
- ❖ Invertebrate Zoology- Robert D. Barnes, Holt-Saunders International Edition, Philadelphia.
- ❖ Biology-Text books for Classes- XI and XII NCERT, New Delhi.
- ❖ Intermediate First year Zoology- 2005, 2008, Telugu Akademi, Hyderabad.
- ❖ Cell and molecular biology 8th ed - De Robertis, B.I Waverly pvt Ltd. New Delhi
- ❖ Biology today and tomorrow. Thomson books - Starr, Evers, Starr
- ❖ Medical physiology 11th edition - Guyton & Hall Elsevier Saunders
- ❖ Human physiology 9th edition. Mc Graw-Hill - Stuart Ira Fox
- ❖ Invertebrate Zoology. R. Chand & company - P.S Dhami & J.K Dhami
- ❖ Biology, 7th ed. Pearson education 2009 - Campbell & Reece
- ❖ Chordate zoology. R. Chand & co.New Delhi - P.S Dhami & J.KDhami
- ❖ Modern text book of zoology, vertebrates. Rastogi publications - R.L Kotpal
- ❖ Introduction to general zoology. New central book agency - Chaki, Kundu & Sarkar
- ❖ Biology 5th ed. Mc Graw -Hill - Peter H. Raven, George B.Johnson
- ❖ Integrated principles of zoology 13th ed. Mc graw-Hill - Hickman,Robert, Larson, I'Anson's
- ❖ The fresh and salt water fishes. Greenwich house newyork - Migdalski, Fichter
- ❖ Animal Ecology – Charles Elton – Macmillan Company 1927
- ❖ Medical Parasitology by Abhay R. Satoskar
- ❖ Human Parasitology, 3rd Edition by Burton J. Bogitsh
- ❖ Essentials of Human Parasitology by Judith Stephenson Heelan
- ❖ Modern Text Book of Zoology, Invertebrates by Kotpal R.L.
- ❖ General Parasitology by Thomas C Chang
- ❖ Drug & Alcohol abuse by Hal Marcovitz
- ❖ Drug and Alcohol abuse - The authoritative guide for Parents, Teachers and Counsellors by H.Thomas Milhorn
- ❖ Ecology Global Insights & Investigations– Peter Stilling – McGraw-Hill
- ❖ Ecology – Begon
- ❖ Encyclopedia of Ecology

WEBSITES

<http://WWW.curiousstaxonomy.net/rules.html>

[http://en.wikipedia.org/wiki/cleavage-\(embryo\)](http://en.wikipedia.org/wiki/cleavage-(embryo))

www.blackwellpublishing.com/ridley/a-z/biological-species-concept.asp

<http://en.wikipedia.org/wiki/species>